

Fix declaration (one page)

Condensed, with no number or claim changed, from the declaration committed before the fix was built (commit 3d54e0d, docs/fix_loop.md sections 1–4). The outcome and post-mortem are in sections 5–8.

1. The worst-performing gate

G-WALL-VIDEO, video wall lengths within $\pm 3\%$: 0 of 23 gate walls pass (bench/results/video_vs_lidar.json, code 1d00698, against the LiDAR plans of the same three walks).

	c00a170fe1	1a8384c3f6	c7d28f72c6	All
Gate walls within 3%	0 of 6	0 of 17	no room paired	0 of 23
Rooms paired with LiDAR	1 of 3	5 of 7	0 of 9	6 of 19
Wall-map agreement with LiDAR	0.47	0.55	0.39	

2. Root cause hypothesis and evidence

Hypothesis. Depth Anything 3 loses track where the phone turns facing a blank wall up close. At some tracking breaks the heading comes out 60–90° wrong, and drift correction can't repair it:

- wall directions only measure heading modulo 90°;
- loop closures search only $\pm 8^\circ$;
- breaks are not declared, so the creep model resists a sudden change.

Evidence. 1. With ARKit's camera path and the same video depth, wall-map agreement rises from 0.39–0.55 to 0.77–0.88. 2. At 13 breaks the median rotation error is 7.2°, but 3 breaks are off by 90.4°, 61.9° and 86.6°. 3. Inside a 12-frame run DA3 is accurate, about 1–2 cm and 1° per frame pair; the damage is done at the joins. 4. On a synthetic walk, a declared 25° break is repaired (43.7 → 0.3 cm), a 62° break snaps to the wrong quarter turn, and a 90° break is left as it was. 5. Ruled out: joining the frames either side of a break in one DA3 run is much worse (71.4° against 7.2°), and denser key frames did not help. Not addressed by this fix: room building on noisy depth (c7d28f72c6's walls read about 15% short even with ARKit's path).

3. The fix

1. Declare tracking breaks (DA3 confidence below 2.0) to drift correction, and hold them loosely.
2. Search every quarter turn for loop closures across a break, and start the pose graph from the turn the loops agree on.
3. Ignore depth from very unsure frames (median confidence below 1.2).

4. Predicted numbers after the fix

	Before	Predicted
Gate walls within 3%	0 of 23	2 (1–4); G-WALL-VIDEO still fails
Paired walls, typical error	15.3%	10% (7–12%)
Rooms paired	6 of 19	9 (8–12)
Agreement c7d28f72c6 / 1a8384c3f6 / c00a170fe1	0.39 / 0.55 / 0.47	0.60 / 0.63 / unchanged ± 0.05

Why it will fall short of the gate: room building on video depth remains, breaks the walk never revisits have no loop to close, and position errors across breaks are corrected only where loops exist.